Learnable Loss Balancing in Anchor-Free Oriented Detectors for Aerial Object

Kai Wang, Zhifeng Xiao, *Member, IEEE*, Qiao Wan, Xiaowei Tan, and Deren Li

***Abstract*— Oriented object detection plays an important role in aerial image interpretation. Image processing speed is also essential due to massive amounts of aerial images. Anchor-free oriented detectors with fast processing speed are generally accepted despite the absence of preset anchors, contributing to their performance gap with anchor-based detectors. Most anchor-free oriented detectors are carefully designed by defining samples according to target characteristics, which require substantial prior knowledge, to realize improved performance. This study proposes an anchor-free oriented detector (termed rfpoint) that requires minimal prior knowledge. Moreover, this study mainly aims to introduce a dynamic sample definition strategy. This strategy is modeled as a dynamic regulating process, wherein the classification and box regression interact until the model converges. A rotating quality-driven loss (RQDL) and an adaptive-weight box loss (AWBL) are also proposed to realize the aforementioned process. RQDL redefines positive and negative attributes of samples according to the distribution of rotating intersection-over-unit (IoU) between predictions and ground truth. AWBL adjusts the importance degree of candidate samples in the box regression through classification scores. The proposed method is then tested on three mainstream aerial image datasets (DOTA, DIOR, and HRSC2016). Results reveal that the proposed method achieves the best performance compared with other oriented detectors, whose mean average precision (mAP) is 79.92%, 70.88%, and 90.67%. Moreover, the method maintains the inference speed advantage of anchor-free detectors. An effective sample definition method can bridge the performance gap of anchor-free oriented detectors without minimizing inference speed.**

***Index Terms*— Aerial image, anchor-free detector, label assignment, oriented object detection.**

I. Introduction

OBJECT detection in natural images has achieved high performance with the rapid development of computer vision algorithms. Aiming at targets in aerial images, which may be densely arranged in any direction and possess large aspect ratios, many high-performance object detectors for the aerial image are proposed on the basis of mainstream object detectors by detecting objects as rotating rectangular. Meanwhile, similar to the accuracy performance, the processing speed becomes an important evaluation indicator with the increase in high-resolution aerial images.

Many anchor-based detectors rely on a large number of preset anchors to reach high accuracies. These detectors usually define samples through the rotating intersection-over-unit (IoU) between anchors and targets, such as region of interest (RoI) transformer [1], refined rotation RetinaNet detector (R^3Det) [2], and single-shot alignment network (S^2A-Net) [3]. However, numerous anchors may lead to a considerable amount of computation, especially for oriented anchors in oriented object detectors. Therefore, many anchor-free detectors without any preset anchors are proposed to detect aerial targets by redefining the representation of rotating boxes, where representative methods include gliding Vertex [4], axis learning [5], and oriented objects detection network (O2-DNet) [6]. Detectors with dense anchors generally perform better than those without in aerial image detection due to the particularity of objects in aerial images.

However, adaptive training sample selection (ATSS) [7] proved that the performance gap between anchor-based and anchor-free detectors is mainly from how to define positive samples and negative samples regardless of the representation of samples (anchors or points). Nevertheless, a remarkable performance gap still exists between the two types of detectors in aerial image detection tasks mainly due to the following reasons.

1) *Choosing true-positive samples is difficult.* Intersecting point samples (or grid point samples) with small targets or targets with large aspect ratios is complicated, resulting in a few positive samples in the training process.
2) *Eliminating false-positive samples is complicated.* Relaxing definition restrictions of positive samples may introduce a few false-positive samples, influencing the training of true-positive samples.

Herein, true-positive samples represent samples defined as positive samples, which are necessary in model training. Meanwhile, false-positive samples, which are falsely defined as positive samples, represent those that may seriously interfere with model training. These problems widely exist in anchor-based and anchor-free detectors, which mostly influence anchor-free oriented detectors. Some investigations on location quality estimation and label assignments have attempted to solve the two problems. The former suppresses the scores of samples with low-quality locations to eliminate potential negative samples, while the latter carefully designs sample definition strategies to identify positive samples. Related representative works include FreeAnchor [8],

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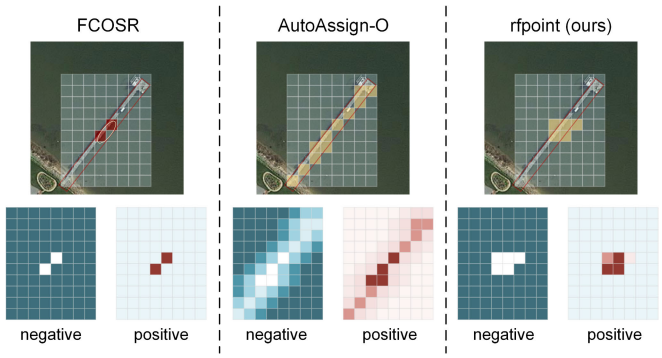


Fig. 1. Visualization of three label assignment strategies. These methods are all anchor-free detectors. AutoAssign-O is the rotating implementation of AutoAssign. Yellow grids donate candidate areas for one scale, and the situation of other scales is similar. Blue and red grids represent negative and positive samples, respectively, and the depth of the color indicates the degree of negative and positive.

generalized focal loss (GFL) [9], AutoAssign [10], self-adaptive anchor selection detector (A^2S-Det) [11], and dynamic anchor learning (DAL) [12].

Most of these investigations focus on anchor-based detectors and follow similar rules, in which positive anchors converge around the center of targets. High-confidence predictions with low-quality locations can be filtered in the nonmaximum suppression (NMS) process based on the aforementioned rule. However, in anchor-free oriented detectors, samples converging around the center is a strict condition, and a well-designed sample definition strategy may not fit all data distribution. Many anchor-free methods define positive samples inside bounding boxes and adopt a centerness branch (or other forms) to filter out low-quality predictions (e.g., fully convolutional one-stage object detector (FCOS) [13], axis learning [5], and general Gaussian heatmap label assignment (GGHL) [14]). But, the centerness branch brings extra computation and contains much prior knowledge. FCOS rotation (FCOSR) [15] and AutoAssign [10] are recent investigations, seeming to handle above problems well. But, there are still some limitations. Fig. 1 shows three kinds of label assignment strategies toward anchor-free oriented detectors corresponding to FCOSR [15], AutoAssign [10], and the proposed method. FCOSR [15] defines samples through a well-designed ellipse in the target and controls the ellipse size to ensure the quality of the samples. This strict design may lead to feature mismatching of the corresponding level, wherein suitable level samples are filtered out due to outside the target. It adopts several rules to control feature level assignment. AutoAssign [10] adopts a Gaussian learning module to learn the shape information of each category, which can gradually reduce the sample definition scope. This module may lead to poor performance in aerial scenes due to some aerial targets with inconsistent shapes. Therefore, a universal and flexible label assignment strategy is essential in anchor-free oriented detectors.

A method that can automatically regulate the factors causing the above problems and only needs a fuzzy positive sample set is proposed to solve these problems and establish a high-performance detector for aerial targets. The proposed method mainly comprises two parts: rotating quality-driven classification regression loss and adaptive-weight box regression loss. First, a set of candidate grid points are selected by distance for each feature level. The candidate set has no additional constraints, such as grid points inside targets or grid boxes intersecting with targets. The candidate samples are all initially treated as negative samples in the classification regression process and then as positive samples in the box regression process. Original local equilibriums are broken with the regression process. Some of the candidate points are gradually assigned as positive samples from negative samples, because precise box regression causes changes in goals of the classification regression loss. Similarly, the classification regression process contributes to the polarization of weights of these candidate points toward 0 and 1, allowing the box regression to focus on points with large weight. The contributions of this article can be summarized as follows.

1) A dynamic sample definition strategy toward anchor-free oriented detectors requiring minimal prior knowledge is proposed. The strategy is modeled as a dynamic regulating process, wherein two core branches of classification and box interact until model convergence.
2) A rotating quality-driven loss (RQDL) and an adaptive-weight box loss (AWBL) are further introduced to achieve the aforementioned process. RQDL adjusts the weights of training samples through quality of predictions and adopts an IoU-based mapping function to address the low scores in targets with large aspect ratios. AWBL distinguishes important samples through an adaptive-weight function related to classification scores.
3) Adequate experiments and detailed analysis on mainstream datasets prove the effectiveness of the proposed method. Learnable loss balancing can well handle the problems of fuzzy sample definition in aerial detection with minimal prior knowledge. Overall, the proposed method achieves the best inference speed while maintaining high performance.

II. Related Works

Aerial image detectors proposed in recent years and studies regarding label assignment in detectors are briefly reviewed in this section.

A. Oriented Object Detection

As a classic task in aerial image detection, current oriented object detectors have achieved remarkable success. With preset dense anchors, anchor-based oriented detectors [1], [16] detect aerial image objects as rotating rectangular. Compared with horizontal boxes, additional angle prediction is adopted in these rotating target detectors. Despite their success, some challenges with discontinuous boundaries in angle prediction are encountered. Circular smooth label (CSL) [17] transforms the task of angle predictions into angle classifications and proposes a CSL method to solve the problems of angle periodicity and error tolerance to adjacent angles. Adaptive period embedding (APE) [18] abandons the routine method of directly predicting angles and directly predicts the trigonometric functions of angles, which can be decoded into angles in return. Furthermore, these angle prediction methods do not substantially rely on preset rotating anchors and can also be effectively applied in anchor-free detectors.

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Similar to research on IoU loss [19] and generalized IoU (GIoU) loss [20], related studies in rotating detectors are also available. These studies describe the relationship between targets and predictions through approximate methods of rotating IoU, which can be directly optimized, due to the nondifferentiable property of rotating IoU. Pixels IoU (PIoU) [21] simulates the rotating IoU using discrete pixels and formulates it into a differentiable form according to a kernel function. Gaussian Wasserstein distance (GWD) [22] represents the rotating boxes as a Gaussian distribution and replaces the rotating IoU with the Wasserstein distance between two Gaussian distributions.

Compared with horizontal detectors, dense anchors, especially rotating anchors in oriented detectors, will introduce a considerable amount of computation. Some researchers focus on the definition of arbitrary quadrilateral targets as an attempt to solve the aforementioned problem. Axis learning [5] detects rotating targets by predicting the axes. Gliding vertex [4] formulates aerial targets as gliding vertices on horizontal boxes, which can represent arbitrary quadrilaterals. CenterMap [23] transforms oriented target prediction to a simple center-probability-map-prediction problem and effectively eliminates the ambiguities between target definitions and background. S^2A-Net [3] proposed a feature alignment module whose anchor refinement network can generate a few high-quality anchors instead of presetting many anchors. Gaussian-mask rotation ship detector (GRS-Det) [24] describes ship targets based on their geometry characteristics through a rotation Gaussian mask module. Foreground refinement detector (FoRDet) [25] combines foreground with contextual representations and helps the detector discriminate foreground regions on feature maps.

B. Label Assignment

Detectors usually adopt a standard rule to distinguish between positive and negative samples in the training process, which may substantially affect the performance. Therefore, some works aim at achieving automatic label assignment. Prime sample attention (PISA) [26] indicates that the prime samples contribute more compared with hard samples and proposes an efficient sampling learning strategy to direct the focus of the training process toward prime samples. ATSS [7] designs a dynamic IoU threshold to define samples whose main contribution lies in proving the importance of defining samples for the performance of detectors regardless of regression by anchors or by points. FreeAnchor [8] represents the regression problem as a maximum likelihood estimation problem, wherein the most representative anchors can be selected by the mean-max function. Inspired by FreeAnchor [8], AutoAssign [10] further reduces prior knowledge, limiting the potential positive sample points inside the ground truth. Probabilistic anchor assignment (PAA) [27] formulates the distribution of anchors as a 2-D Gaussian mixture model, which can be solved by the expectation maximization algorithm. However, the solution procedure consumes a considerable amount of time. To solve the ambiguous assignment of overlapping samples, optimal transport assignment (OTA) [28] models the label assignment process as an optimal transport problem and adopts Sinkhorn–Knopp iteration to solve this optimization quickly. RFLA [29] proposes a receptive field distance (RFD) based on Wasserstein distance and Kullback–Leibler divergence (KLD) to measure the similarity between Gaussian receptive field and ground truth. Positive samples are then chosen in accordance with top-k RFD scores.

The general sample definition method cannot work effectively when applied to aerial image detection, especially for objects with large aspect ratios. Some research regarding label assignment in aerial image detection is also available. A^2S-Det [11] maintains a balance between positive and negative anchors from a set of candidate anchors selected by horizontal IoU. Considering the quality of preset anchors and predicting boxes synthetically, group assignment strategy (GAS) [30] splits anchors into several groups according to the IoU between anchors and ground truth and assigns anchors by groups. DAL [12] proposes a DAL method, which establishes a newly defined matching degree to evaluate the potential for each anchor. FCOSR [15] employs ellipse areas to constrain positive samples and design a fuzzy sample label assignment strategy to solve assignment challenges toward different feature levels. Anchor-free detector with feature enhancement and soft label assignment (FSDet) [31] proposes a soft label assignment module to measure the importance of training samples and focus on these important samples in the training process. GGHL [14] designs a GGHL strategy to adjust the Gaussian center prior weights adaptively to fit the characteristics of different objects.

Aforementioned investigations can be summarized as strategy-based approaches [7], [11], [12], [26], [27], [28], [29], [30] and dynamic-loss-based approaches [8], [10], [14], [15], [31]. The former defines positive and negative samples through hand-designed strategies, while the latter employs dynamic loss functions to achieve dynamic and adaptive label assignment. The proposed method belongs to dynamic-loss-based approaches.

III. Proposed Method

For the detector structures, FCOS [13] is mainly used as a reference, and the backbone, feature pyramid network (FPN), and detector heads are adopted as basic structures. Considering the difference in label assignment strategy, the centerness head is dropped out, and only the classification and box prediction heads are retrained. A few changes in structures prompt the transformation from horizontal detection to oriented detection. The current work focuses on label assignment in the anchor-free detector, which is reflected in the design of losses. Similar to most studies on label assignment, the presented improvements only slightly increase the computation amount of the training process instead of the inference process.

A. Overview

The proposed method abstracts the classification and box regression process as a regulating process. As shown in Fig. 2, the network outputs two heads to predict categories and rotating boxes. Classification and box regression losses,

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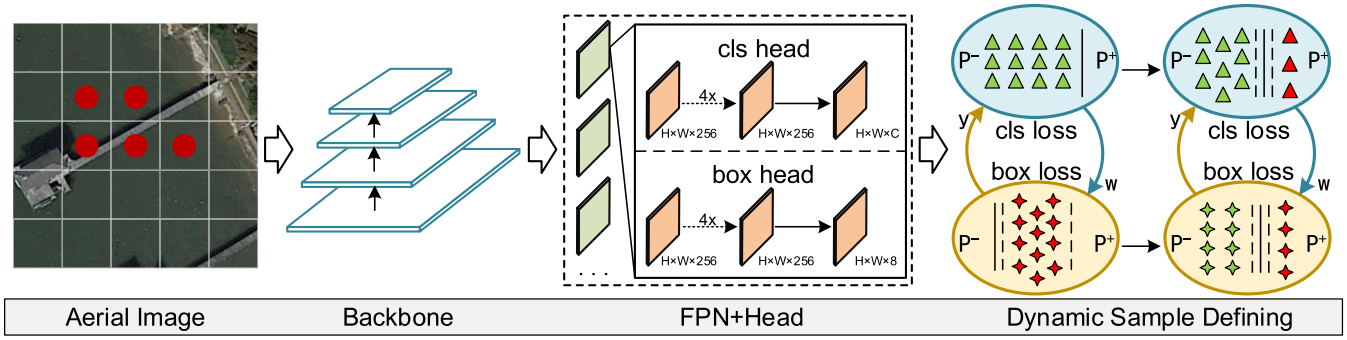


Fig. 2. Architecture of the proposed method. First, the network comprises backbone, FPN, and detector heads. The backbone donates C_3–C_5 layers, and the FPN donates P_3–P_7 layers. Second, the approach defining positive and negative samples is based on grid points, which is similar to approaches of most anchor-free detectors. Third, the dynamic sample definition approach is the core contribution of the proposed method, which can be formulated as two branch losses. Fourth, P^- and P^+, respectively, represent sets of negative and positive samples in the training process. In the original phase, all candidate samples are assigned as positive samples (red symbol) and negative samples (green symbol) in the box and classification branches, respectively. With the interaction of the two branches, some candidate samples are gradually redefined as positive or negative samples in the training process.

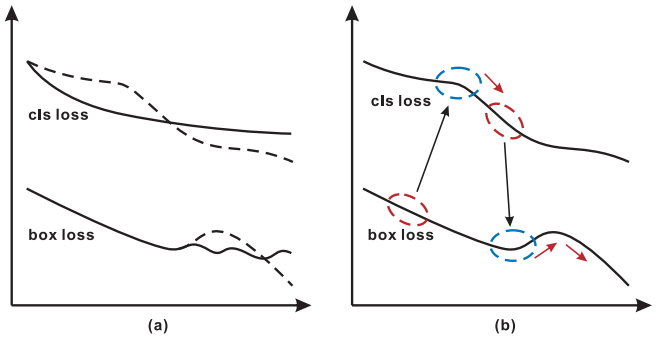


Fig. 3. (a) and (b) Loss optimization diagrams of the general label assignment method and the proposed method, respectively. (b) Interaction of the two branches and their effects on respective optimization processes. In addition, red ovals represent qualitative change due to the quantitative change. Blue ovals represent regions where the optimization curve demonstrates sudden changes. Red arrows represent the optimization direction after sudden changes in the curve.

which regulate reciprocally in training, are designed on the basis of the two heads. First, a set of candidate points are chosen by distances where each scale of layers donates the nearest k points. The distance represents Euclidean distance between the center point of objects and point samples in feature maps. At the beginning of the training process, all the candidate points are assigned as negative samples (P^-) calculating classification loss, while they are equally weighted to calculate box regression loss as positive samples (P^+). As the training progresses, the quantitative change in the box regression branch causes the qualitative change in the classification branch, wherein some of the candidate points are gradually assigned as positive samples (P^+) in the classification branch. Meanwhile, the change in the classification branch also influences the box regression branch, wherein candidate points of equal importance are divided into positive samples (P^+) and negative samples (P^-) in the box regression branch. The two branches interact similar to positive feedback regulation of biology until the end of the training. As shown in Fig. 3, from the perspective of optimization, the two curves break the local optimum to find the optimal under the influence of each other.

B. Network Architecture

Using ResNet as the backbone to extract rich feature information from images, the following FPN is mainly in charge of multiscale feature fusion and multilayer detection, which donates P_3–P_7 layers. Furthermore, two subnetworks are used to predict categories of targets and the encoded vectors of bounding boxes for each P_i. In most anchor-based rotating detectors, box encoding methods depend on numerous preset anchors, where dense anchors facilitate the regression of (x, y, w, h, θ), especially for the angle θ. APE encoding methods, which encode rotating bounding boxes as shown below, are directly introduced to bridge this gap

$$t_x = \frac{x - x_p}{s_p} \quad t_y = \frac{y - y_p}{s_p}$$
$$t_w = \ln\left(\frac{w}{s_p}\right) \quad t_h = \ln\left(\frac{h}{s_p}\right)$$
$$(u_1, u_2, v) = \mathrm{APE}(w, h, \theta) \tag{1}$$

where (x, y, w, h, θ) denotes bounding box of the target, while (x_p, y_p, s_p) denotes coordinates(x_p, y_p) and downsampling ratio (s_p) of the corresponding grid point. (u_1, u_2, v) are the values of angles encoded by APE encoding, where u_1 represents $(\cos 4\theta, \sin 4\theta)$ and u_2 represents $(\cos 2\theta, \sin 2\theta)$. v is the scale coefficient when calculating smooth$_{L1}$ of the item u_2.

The category prediction subnetwork outputs a final layer with the shape of (N, W, H, C), which denotes the batch size, the size of P_i layer, and the number of categories. Moreover, the bounding box prediction subnetwork predicts a tuple of $(t_x, t_y, t_w, t_h, \cos 4\theta, \sin 4\theta, \cos 2\theta, \sin 2\theta)$ whose final layer has the shape of $(N, W, H, 8)$.

C. Rotating Quality-Driven Loss

Positive training sample set roughly chosen by distance reveals the need to suppress some false-positive samples, while others are true-positive samples benefiting to the detector training. Quality focal loss (QFL) [9] is an effective loss to relate the classification regression process with the box quality of predictions. However, application in aerial scenes has many limitations. On the one hand, rotating IoU is a

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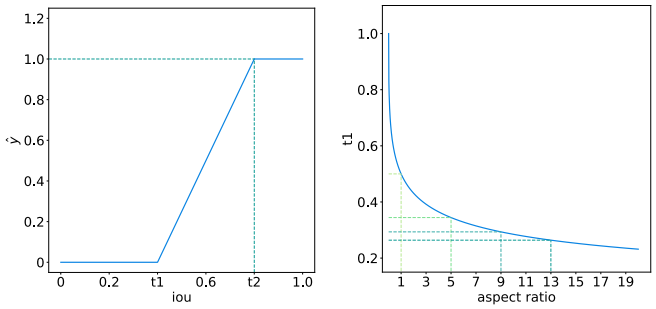


Fig. 4. Classification regression goals ($\hat{y}$) depend on the distribution of candidate samples and the negative threshold. (a) Visualizations of the mapping function $f(\text{IoU})$. (b) Visualizations of the dynamic threshold function $t_1(r)$.

strict evaluation method compared with natural scenes. For aerial targets with large aspect ratios, rotating IoU between predictions and ground truth is probably at a low level even in the late training process. Therefore, QFL suppresses scores of true-positive samples, influencing the final performance. On the other hand, false-positive samples may cause phenomena where the degree of classification regression is always lower than the box regression toward these samples.

RQDL is proposed to solve the above challenges. The core idea is to relate classification scores with the quality of box predictions. Therefore, $(y-\sigma)$ in focal loss (FL) is replaced with $(\tilde{y}-\sigma)$. Herein, σ and y, respectively, represent the classification score and regression goal. y in FL is derived from the one-hot encoding of labels whose value is 0 or 1. Rotating IoUs between targets and predictions are adopted as regression goal $\tilde{y}$ whose values range from 0 to 1 to relate scores with the quality of box predictions. The formulation can be represented as follows:

$$\text{Loss}^{-}_{\{\text{cls}\}} = -|\tilde{y}-\sigma|^{\gamma}\log\big(1-|\tilde{y}-\sigma|\big). \tag{2}$$

A mapping function $f(\tilde{y})$ is designed to replace $\tilde{y}$ as shown in the formulations below to weaken the influence of targets with large aspect ratio. Herein, $\tilde{y}_{(i,j)}$ donates the rotating IoU between the ith target and the jth sample. $\tilde{y}_{\max_i}$ represents the max rotating IoU between the ith target and all samples. Negative threshold $t_1(r_i)$ is adopted to suppress samples with poor box predictions, which changes with the aspect ratios r_i of the ith target. As shown in Fig. 4(a), the mapping function $f(\tilde{y})$ ensures at least one positive sample for training unless no sample reaches the negative threshold. Meanwhile, the function also ensures that all max values of $f(\tilde{y}_{(i,j)})$ can reach 1, avoiding low scores of targets with large aspect ratio

$$f\big(\tilde{y}_{(i,j)}\big) = \begin{cases} \dfrac{\tilde{y}_{(i,j)} - t_1(r_i)}{\tilde{y}_{\max_i} - t_1(r_i)}, & \tilde{y}_{(i,j)} \geq t_1(r_i) \\ 0, & \tilde{y}_{(i,j)} < t_1(r_i) \end{cases} \tag{3}$$

$$t_1(r_i) = \frac{1}{1+r_i^{\lambda}}. \tag{4}$$

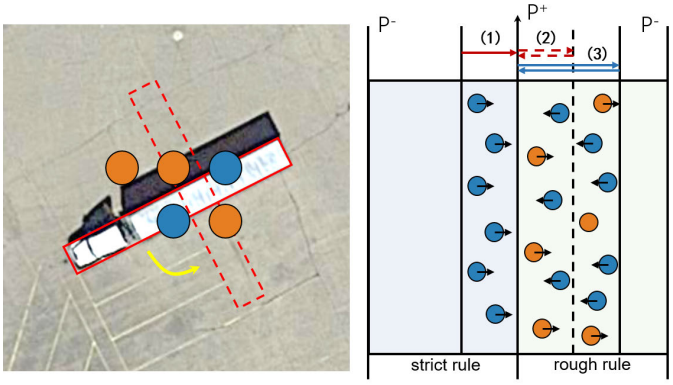


Fig. 5. P^{+} and P^{-} indicate boundaries of negative and positive samples, respectively. General label assignment strategy is tagged as the strict rule. Rough label assignment strategy in the proposed method is tagged as the fuzzy rule. (1)–(3) separately represent modulating factors of $|y-\sigma|^{\gamma}$, $|\tilde{y}-\sigma|^{\gamma}$, and 1. The arrow interval indicates the range of regression capability.

$\tilde{y}$ in formula (2) is replaced with $f(\tilde{y}_{(i,j)})$ and term as $\hat{y}_{(i,j)}$. Therefore, RQDL can be transformed as follows:

$$\text{Loss}_{\{\text{cls}\}} = -|\hat{y}-\sigma|^{\gamma}\log\big(1-|\hat{y}-\sigma|\big). \tag{5}$$

Nonetheless, high-score predictions with poor localization exist, because some samples are far away from the center of ground truth in such a fuzzy candidate set. Moreover, limitations of feature extraction networks cause insufficient information of these samples to suppress the classification scores. The same challenge also exists in dense anchor-based detectors. Candidate samples are generally chosen in dense anchors according to rotating IoU, which are most likely near the center of the ground truth. Only a few low-quality samples are observed in this situation, and most predictions of low-quality samples may be filtered out by the NMS process. Formula (5) considered this problem. $\hat{y}$ is correlated with rotating IoU, representing the rotating localization quality of predictions. Regression value $|\hat{y}-\sigma|$ will treat samples as negative samples when σ is substantially higher than $\hat{y}$. However, the modulating factor $|\hat{y}-\sigma|^{\gamma}$ may weaken this process whose original motivation is to direct the focus of the detector on positive samples. Fig. 5 shows the regression capability differences of three modulating factors. $|y-\sigma|^{\gamma}$ and $|\tilde{y}-\sigma|^{\gamma}$ work well in FL and QFL, respectively, because most samples are defined appropriately. However, fuzzy candidate set causes the uncertainty of sample definition (as shown in Fig. 5). This uncertainty must be learned by the network. $|\hat{y}-\sigma|^{\gamma}$ reduces the gradient of this process and suppresses network learning of the uncertainty of sample definition. Therefore, the modulating factor $|\hat{y}-\sigma|^{\gamma}$ is eliminated to reinforce this process when samples are candidates. The final RQDL can be represented as follows:

$$\text{Loss}_{\{\text{cls}\}} = \begin{cases} -\log\big(1-|\hat{y}-\sigma|\big), & \text{candidate}^{+} \\ -|\hat{y}-\sigma|^{\gamma}\log\big(1-|\hat{y}-\sigma|\big), & \text{others}^{-}. \end{cases} \tag{6}$$

The focus term $|y-\sigma|^{\gamma}$ in FL is to make detectors focus more on hard samples. Here, $\hat{y}$ in $|\hat{y}-\sigma|^{\gamma}$ is related with qualities of box predictions. It helps regulate important degrees

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of simple samples and hard samples, playing a focus-term role in RQDL. Thus, directly removing $|\hat{y} - \sigma|^{\gamma}$ does not affect the focusing process in the model training.

D. Adaptive-Weight Box Loss

In general box regressions, all positive samples are of equal importance to the box regression loss. However, defining positive samples is difficult, and distinctions of weights between these positive samples should be realized. All candidate samples are treated in this section as negative samples in the classification regression process until local equilibrium is broken by gradual accurate prediction boxes. Box regression loss is defined in a similar way. However, if the same candidate samples are also treated as negative samples in the box regression process at the beginning, then the two branches will be in respective local equilibrium and will wait for each other to break. Therefore, an AWBL mainly related to classification scores is designed to distinguish important samples in the box regression process; thus, the regression process provides additional attention to important samples. Opposite to the loss in Section III-C, all positive samples are treated as positive samples with equal weights at the beginning and weights adaptively change with classification scores. First, the box regression item can be described as follows:

$$\begin{aligned}&\text{smooth}_{L1}\{\text{APE}\}\\&= \sum_{t_x,t_y,t_w,t_h,u_1} \text{smooth}_{L1}(\Delta t) + \sum_{u_2} v \cdot \text{smooth}_{L1}(\Delta u)\end{aligned} \tag{7}$$

where $(t_x, t_y, t_w, t_h, u_1, u_2)$ is the vector to regress, representing the offsets relative to the corresponding points. In the formula, $\text{smooth}_{L1}(\Delta)$ is the loss generally used in box regression, which is

$$\text{smooth}_{L1}(\Delta) = \begin{cases} 0.5(\Delta)^2, & \text{if } |\Delta| < \beta, \\ (|\Delta| - 0.5\beta), & \text{others.} \end{cases} \tag{8}$$

The box regression term is represented as probability, which can be formulated as shown below, to normalize the loss value

$$p_{(i,j)} = e^{-\text{smooth}_{L1}\{\text{APE}\}}. \tag{9}$$

Samples with high confidence should provide additional attention to the box regression process. Hence, $p_{(i,j)}$ is weighted by $w_{(i,j)}$ when calculating box losses

$$g(\varphi) = \frac{1}{1-\varphi} \tag{10}$$

$$w_{(i,j)} = \frac{g\left(x_{(i,j)} \cdot \text{mask}_{(i,j)}\right)}{\sum_j^{M_{\text{top-}k}} g\left(x_{(i,j)} \cdot \text{mask}_{(i,j)}\right)}. \tag{11}$$

$g(\varphi)$ is a common mapping function, mapping φ from (0, 1) to $(1, \infty)$. $x_{(i,j)}$ denotes the classification scores of the jth point sample toward the ith target. Targets in the aerial image are generally densely arranged, and the same candidate point may belong to several different targets, which may cause fake weights. As shown in Fig. 6, the same ambiguous point sample toward the two targets both reach high classification scores, and the prediction box only belongs to one target. The same point sample belonging to several targets and with the same class cause the inconsistency between localization and classification. Therefore, $\text{mask}_{(i,j)}$ is adopted on the basis of IoU to restrict the attribution of weights, where

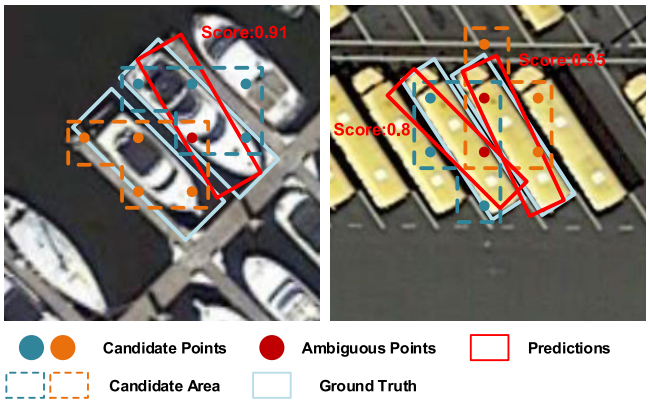


Fig. 6. As shown in the above figure, red points belong to two targets of the same category, and red rectangles are predictions of corresponding points. Ambiguous points are assigned to corresponding targets based on the gradual prediction precision. However, for the other targets, predictions of these ambiguous points are of high confidence and poor localization. If the box regression branch is only driven by classification, then these ambiguous points will lead to high fake weights of other targets, seriously affecting the box regression branch. Therefore, the localization factor $\text{mask}_{(i,j)}$ is introduced as a weak positioning constraint.

$$\text{mask}_{(i,j)} = \frac{g\left(\text{iou}_{(i,j)}\right)}{\max_j g\left(\text{iou}_{(i,j)}\right)}. \tag{12}$$

In formula (12), $\text{iou}_{(i,j)}$ represents the iou between the jth point sample and the ith target. The function $\text{mask}_{(i,j)}(\text{iou}_{(i,j)})$ allocates the point samples to different targets according to the weighting method. $\text{mask}_{(i,j)}$ fluctuates around 1 when the IoU of candidate samples is all at a low level, indicating that the point sample is equal toward all corresponding targets. As the training continues, the distribution of $\text{mask}_{(i,j)}$ turns into polarization from smoothness, while the point sample gradually belongs to one target. Combining the above discussion, box regression items $p_{(i,j)}$ are summed weighted by $w_{(i,j)}$. The final AWBL can be formulated as follows:

$$\text{Loss}_{\{\text{box}\}} = \sum_i^{N_{\text{gt}}} \left(\log \left(1 - \sum_j^{M_{\text{top-}k}} \left(w_{(i,j)} \cdot p_{(i,j)} \right) \right) \right). \tag{13}$$

$M_{\text{top-}k}$ and N_{gt} are the number of candidate point samples and targets, respectively.

E. Loss Function

In Sections III-C and III-D, classification loss $\text{Loss}_{\{\text{cls}\}}$ and box regression loss $\text{Loss}_{\{\text{box}\}}$, which form the total loss, are specially designed. The loss summed for all samples must be divided by a coefficient, which is generally represented by the number of positive samples. The coefficient has substantial influence, and its determination of the value becomes an established rule [8], [32]. Following the previous study, every loss must be divided by the number of positive samples selected ($N_{\text{gt}} \cdot \text{top-}k$). Summed weighted by $w_{(I,j)}$ (where $\sum_j w_{(i,j)} = 1$), the box regression item $\text{Loss}_{\{\text{box}\}}$ must only

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be divided by the number of targets N_{gt}. The total loss can then be formulated as follows:

$$\text{Loss} = \frac{1}{N_{gt}} \cdot \sum \text{Loss}_{\{box\}} + \frac{1}{N_{gt} \cdot \text{top-}k} \cdot \sum \text{Loss}_{\{cls\}}. \quad (14)$$

IV. Experiments and Analysis

Experiments on several aerial image rotating object detection datasets DOTA [33], DIOR [34], and HRSC2016 [35] are conducted to verify the universality of the proposed method. Sections IV-A–IV-D provide the details of datasets, implementation, results, and discussions.

A. Datasets

1) DOTA: This dataset contains 2806 aerial images with 188 282 annotated instances of 15 categories whose image size ranges from around 800 × 800 to 4000 × 4000 pixels. The categories of DOTA include the following: plane, baseball diamond, bridge, ground track field (GTF), small vehicle, large vehicle (LV), ship, tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field, roundabout (RA), harbor, swimming pool (SP), and helicopter (HC). The dataset is divided into three parts of training, validation, and testing by the publisher of DOTA. Training some images with large image sizes is difficult; thus, images are cropped into 1024 × 1024 patches with the overlap of 200 pixels, which are suitable as the input of the network. Removing image patches without instance, a total of 12 800 image patches are available for training and validation. For multiscale experiments, origin images will be resized at three scales (0.5, 1.0, and 1.5) and cropped into 1024 × 1024 patches with an overlap of 500 pixels.

2) DIOR: This dataset has 23 463 images with 192 518 instances, totally covering 20 categories, including airplane (APL), airport (APO), baseball field, BC, bridge (BR), chimney (CH), dam (DAM), expressway toll station, expressway service area, golf field (GF), GTF, harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), TC, train station, vehicle (VE), and windmill (WM). Except for common categories similar to most aerial image datasets, some categories of targets without distinct boundaries, such as airport and GF, increase the complexity of detection tasks. The entire dataset is officially split into two sets of train-val and testing, respectively, including 11 725 and 11 738 images. Furthermore, the image sizes are all 800 × 800 pixels.

3) HRSC2016: HRSC2016 is a ship detection dataset including 436 training images, 181 validation images, and 444 testing images. All images are padded as square-sized images, subject to the long side. Images are resized to 800 × 800 pixels.

If not specifically indicated, then all experiments involve training on the union set of training and validation and testing on the testing set. The input sizes of the network are the same as image sizes after resizing.

B. Implementation Details

The proposed method is implemented with MMRotate [36] and mainly refers to several official implementations of mainstream methods, such as FCOS and FreeAnchor. Stochastic gradient descent optimizer is adopted to train the model whose momentum and weight decay are set to 0.0001 and 0.9, respectively. The initial learning rate is set to 0.01. The learning rate decays to 0.1 times the current learning rate at each decay step, which is set to {8/12, 11/12} of total epochs; 36 epochs, 48 epochs, and 72 epochs are run for DOTA, DIOR, and HRSC2016, respectively. Furthermore, the batch size of model training and inference is separately set to 8 and 1, respectively. Random data augmentation, including random mirror flip and random rotating, is adopted in the training process to facilitate fair comparison with other methods.

C. Ablation Studies

1) Effectiveness of Dynamic Sample Definition: Section I discussed that sample definition strategy differences contribute to the performance gap between anchor-based and anchor-free oriented detectors. The proposed method defines a fuzzy set of training samples according to distance and achieves dynamic sample definition through losses introduced in Sections III-C and III-D. Unlike other sample definition strategies, the proposed method does not require additional constraints (e.g., grid point samples must be inside the targets). This characteristic helps the detector need no extra design and parameter toward sample definition, especially when facing other unknown training data. Table I shows ablation experiments on RQDL and AWBL. The baseline is rfpoint with only QFL. Moreover, different backbones are adopted to prove the robustness of the proposed modules. Table I reveals that the proposed methods achieve increments in average precision (AP) on these datasets. For the backbone of ResNet50 (R50), 1.44% and 1.56% increments of AP in the DOTA dataset are observed when adopting RQDL and AWBL, respectively. Similar increments, which are 2.73% and 2.31%, also exist in the DIOR dataset. However, ambiguous increments are observed in the HRSC2016 dataset. The scale of the HRSC2016 dataset is substantially smaller than DOTA and DIOR. Current methods focus on the upper limit of the HRSC2016 dataset, and observing increments is difficult. The backbone of Swin-Transformer-Tiny (Swin-t) also reveals increments of AP in these datasets. These ablation experiments can realistically prove the effectiveness and robustness of the proposed method.

2) Effectiveness of Rotating Quality-Driven Loss: Compared with other approaches, the proposed method lost a considerable amount of prior location information due to fuzzy candidate samples. The RQDL proposed in Section III-C successfully solves the interference problem of false-positive samples with training. This condition primarily benefits from the rotating IoU mapping function corresponding to formula (4). Meanwhile, the problem of poor scores can also be solved for some large-aspect-ratio targets, which benefits from the mapping function and the loss form in formula (6). Fig. 7 depicts the distribution of harbor and LV prediction scores when processing the same data. Fig. 7 reveals that the proposed method outputs more high-score and fewer low-score predictions than the baseline. Results in Table II also show the effectiveness of RQDL. Adopting RetinaNet [32] as the baseline, 1.59% increase of mean AP (mAP) is observed

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TABLE I

ABLATION EXPERIMENTS OF THE PROPOSED LOSSES ON THREE MAINSTREAM DATASETS (DOTA, DIOR, AND HRSC2016). CLS BRANCH REPRESENTS THE USE OF ROTATING QUALITY-DRIVEN LOSS. REG BRANCH REPRESENTS THE USE OF AWBL. TRAINING ONLY USES RANDOM MIRROR FLIP AS DATA AUGMENTATION

Methods	Cls Branch	Reg Branch	AP_{DOTA}	AP_{DIOR}	$AP_{HRSC2016}$
rfpoint-R50	×	×	70.47	59.76	90.04
	✓	×	71.91(+1.44)	62.49(+2.73)	**90.27(+0.23)**
	✓	✓	**73.47(+1.56)**	**64.8(+2.31)**	90.21(-0.06)
rfpoint-Swin-t	×	×	73.45	66.12	89.9
	✓	×	74.91(+1.46)	68.68(+2.56)	90(+0.10)
	✓	✓	**75.96(+1.05)**	**70.88(+2.20)**	**90.15(+0.15)**

TABLE II

ABLATION EXPERIMENTS OF ROTATING QUALITY-DRIVEN LOSS ON THE DOTA DATASET. QFL REPRESENTS THE QUALITY FOCAL LOSS. RQDL INDICATES ROTATING QUALITY-DRIVEN LOSS

methods	module	AP_{HA}	AP_{LV}	AP_{BR}	AP_{50}	AP_{75}	AP
RetinaNet-R50	+QFL	53.90	62.56	**41.73**	68.67	39.20	39.13
	+RQDL	**58.98**	**68.55**	40.00	**70.26**	**42.08**	**41.15**
rfpoint-R50	+QFL	62.93	78.83	48.93	71.54	42.06	41.26
	+RQDL	**66.00**	**79.34**	**50.69**	**73.47**	**44.40**	**43.10**

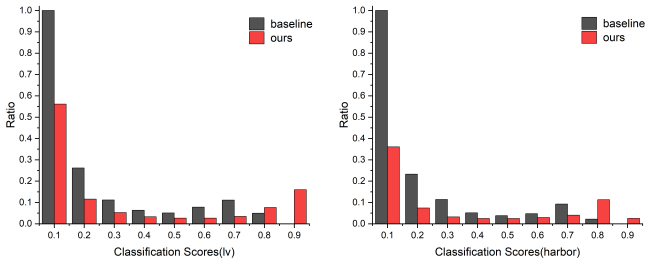
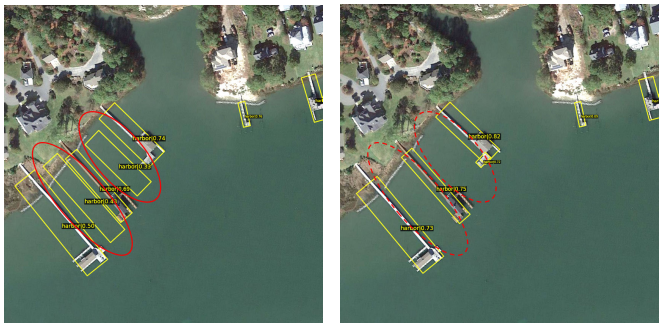


Fig. 7. Statistical histograms of prediction scores for the harbor and LV on the DOTA dataset. The y-axis values reflect predicted target numbers, which are divided by the maximum value for display. The x-axis values reflect scores. The gray blocks represent the baseline method, which employs the origin of QFL. The proposed RQDL is indicated by red blocks.

TABLE III

HYPERPARAMETER EXPERIMENTS ON THE DOTA DATASET FOR THE ROTATING QUALITY-DRIVEN LOSS, INCLUDING RESULTS OF FIXED t_1 AND DYNAMIC t_1. r REPRESENTS THE ASPECT RATIO OF TARGETS

t1	λ	AP_{HA}	AP_{ST}	AP_{50}
$t_1 = 0$	-	67.09	82.20	72.48
$t_1 = 0.5$	-	65.42	83.82	72.82
$t_1 = 0.7$	-	59.92	84.59	68.38
$t_1 = \frac{1}{1+r^{\lambda}}$	0.2	67.35	**85.33**	73.04
	0.4	66.00	84.75	**73.47**
	0.8	**67.64**	83.89	73.03
	1.6	67.22	84.55	72.83



(a) (b)

Fig. 8. Detection visualizations for adopting different modulating factors in the RQDL. (a) Modulating factor $|\hat{y} - \sigma|^{\gamma}$. (b) Discarding the modulating factor. Yellow rectangles represent detection results. (a) Weak modulating factor. (b) Strong modulating factor.

when using RQDL compared with QFL. Similarly, a 1.93% increase in mAP is found when using the introduced network as the baseline. Performance increments are evident especially for large-aspect-ratio targets, such as the harbor (HA), the LV, and the bridge (BR). These results confirm the accuracy of the motivation of researchers, where strict rotating IoU from QFL may restrain the scores of large-aspect-ratio targets. The cutoff value (t_1) relates to the mapping function $f(\text{iou})$, directly influencing the regression tendency of candidate samples. Different t_1 values are beneficial to the detection of aerial targets with different aspect ratios. Table III shows the results with different hyperparameters. Results indicate that fixed t_1 influences performances of targets with different aspect ratios. Low t_1 fits large-aspect-ratio targets, such as harbor (HA) and bridge (BR). High t_1 fits small-aspect-ratio targets, such as ST and plane. Dynamic t_1 varies with the aspect ratio (r) to suit different aerial targets. The hyperparameter λ controls the values of $t_1(r)$. Results in Table III indicate that the proposed detector reaches the best AP_{50} when $\lambda = 0.4$. Fig. 8 shows the result visualizations with different modulating factors, which are mentioned in Section III-C.

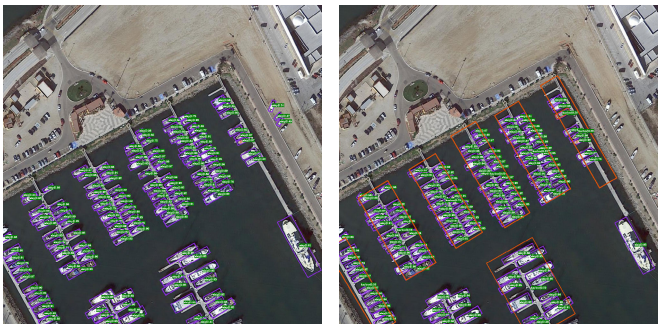
3) Effectiveness of Adaptive-Weight Box Loss: Section III-D discussed the importance of distinguishing weights when training the box branch. AWBL is proposed, increasing the attention of the model to samples with high classification scores. Results in Table I indicate a 1.56% and 1.05% increase of mAP on ResNet50 and Swin-tiny, respectively, when the box branch regresses with weights. The regression process will rapidly focus on samples with high scores when qualitative changes in candidate samples occur in classification scores. This situation will help true-positive samples in the elimination of the influence of false-positive samples, leading to improved performance. Moreover, adaptive weights in the box regression are beneficial to automatic feature selection toward different levels. Common anchor-free oriented detectors assign samples to different feature levels mainly according to sizes. The

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TABLE IV

COMPARISON OF THE SPEED OF THE PROPOSED METHOD ON THE DOTA DATASET WHEN USING DIFFERENT BACKBONES. VALUES CONTAINING * INDICATE THE USE OF FP16 IN THE INFERENCE PROCESS. PP REPRESENTS POSTPROCESSING, WHICH MAINLY INCLUDES THE DECODING AND NMS PROCESSES. ALL STATISTICAL PROCESSES ARE COMPLETED ON A SINGLE V100-16G

Method	Backbone	mAP	Speed (fps) with PP (FP32 / FP16)	Speed (fps) w/o PP (FP32 / FP16)	Params (M)	GFLOPs
rfpoint‡	MobileNetv2	77.39	**39.3 / 43.1**	**48.6 / 53.8**	**7.42**	**101.46**
	R50	77.74	24 / 40	29.3 / 50.1	36.05	208.02
	R101	78.78	19.2 / 31.3	21.2 / 39.2	55.05	285.92
	RX50	78.92	21 / 30.4	23.9 / 38.3	35.54	211.18
	Swin-t	**79.92**	18.5 / 24.3	21.1 / 26.1	36.77	214.18



(a) (b)

Fig. 9. Result visualizations of box regression with weights. (a) Without weights. (b) With weights.

AWBL can achieve automatic feature selection without an effective design, which is advantageous in situations with overlapping targets. Result visualizations in Fig. 9 also prove the effectiveness of the proposed method.

4) Inference Speed and Accuracy: Inference speed has received a considerable amount of attention in aerial image detections due to the high resolution, enormous image size, and large number of aerial images. The proposed method has the fastest inference speed while maintaining good accuracy. Results in Table IV show performance differences in speed and precision with various parameters. The proposed method, which employs MobileNetv2 as its backbone, achieves the fastest inference speed. Postprocessing consumes a fixed portion of the overall process time, with a processing time of around 5 ms on a single V100-16G. Therefore, postprocessing has minimal effect on total processing time considering relatively slow detector speed. However, the postprocessing becomes a bottleneck when the speeds of detectors are fast (e.g., the proposed detector with the backbone of MobileNetv2). From a performance viewpoint, Swin-tiny achieves the best accuracy, which is higher than other backbones. With multiscale training and using abundant data augmentation, large backbones help improve the accuracy of the detector but slow the inference speed. Therefore, choosing the basic network according to the specific requirements of tasks is reasonable to balance speed and precision. Moreover, FP16 inference is a general and effective method to improve inference speed with almost no decrease in accuracy, as shown in Table IV. The theoretical performance of FP16 is approximately twice the FP32 on V100-16G. For backbones of ResNet50 (R50), ResNet101 (R101), and ResNeXt50-32 × 4d (RX50), the proposed method with FP16 inference reaches almost twice the speed than FP32 inference. Even using MobileNetv2 as the backbone, speed improvement is observed with FP16 inference.

TABLE V

SPEED EXPERIMENTS IN DIFFERENT ENVIRONMENTS ON THE DOTA DATASET. THE CONSISTENT MAIN PARAMETERS, INCLUDING BACKBONE, IMAGE RESOLUTION, AND GPU DEVICE, ARE CONTROLLED TO FACILITATE OBJECTIVE COMPARISON WITH OTHER METHODS. ‡ REPRESENTS MULTISCALE TRAINING AND TESTING UNDER RANDOM DATA AUGMENTATION. SPEED VALUE WITH $^-$ INDICATES THAT THE POSTPROCESSING TIME IS IGNORED AND ONLY INVOLVES THE INFERENCE TIME OF THE NETWORK

Method	Backbone	Resolution	mAP	Speed	Device
FCOSR‡ [15]	MobileNetv2	1024x1024	76.11	23.7	2080Ti-11G
rfpoint‡	MobileNetv2	1024x1024	**77.39**	**27.9**	2080Ti-11G
FCOSR‡ [15]	RX50	1024x1024	**79.25**	14.6	2080Ti-11G
rfpoint‡	RX50	1024x1024	78.92	**17.2**	2080Ti-11G
S^2A-Net [3]	R50	1024x1024	**74.12**	16.00	V100-16G
rfpoint	R50	1024x1024	73.47	**24.00**	V100-16G
S^2A-Net‡ [3]	R50	1024x1024	**79.42**	16.00	V100-16G
rfpoint‡	R50	1024x1024	77.74	**24.00**	V100-16G
GGHL‡ [14]	MobileNetv2	800x800	71.26	62.07^-	3090-24G
rfpoint‡	MobileNetv2	800x800	**73.20**	$\mathbf{87}^-$	3090-24G

Backbones, image resolutions, GPU devices, and statistical approaches are important factors influencing performance evaluations, especially for speed. Several experiments with similar environments are designed to compare the proposed method with other mainstream approaches to speed and precision objectively. As shown in Table V, the proposed method achieves the best speed while maintaining competitive precision. GGHL [14], FCOSR [15], and S^2A-Net [3] are recent mainstream one-stage aerial detectors, which maintain a good balance between speed and precision. Employing image resolution of 1024 × 1024, FCOSR‡ [15] achieves the mAP of 76.11% with the inference speed of 23.7 frames/s and the mAP of 79.25% with the inference speed of 14.6 frames/s on a single 2080Ti-11G when MobileNetv2 and RX50 are, respectively, used as backbones. Employing image resolution of 800 × 800, GGHL‡ [14] achieves the mAP of 71.26% with the inference speed of 62.07 frames/s on a single 3090-24G. The proposed method reaches the mAP of 77.39% with a speed of 27.9 frames/s, the mAP of 79.25% with a speed

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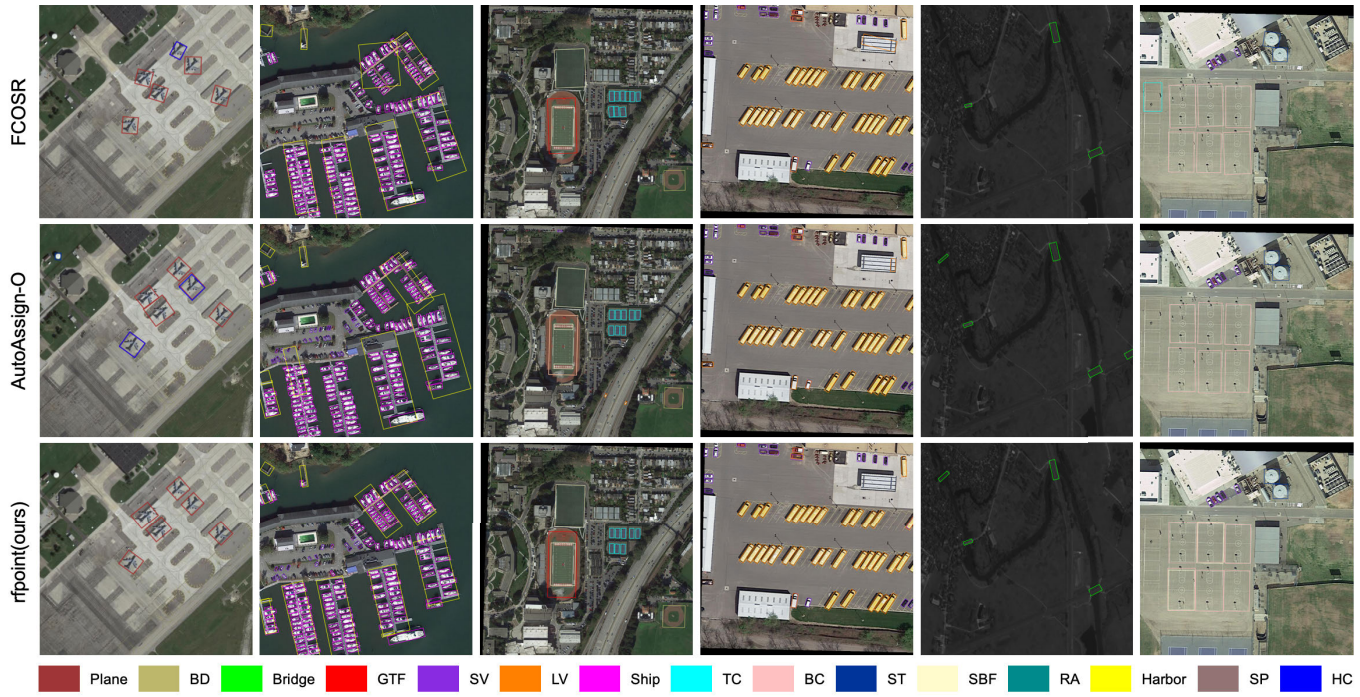


Fig. 10. Visualizations of predictions separately using FCOSR, AutoAssign-O, and the proposed method on the DOTA dataset. Models are trained with basic data augmentation. AutoAssign-O is the rotating implementation of AutoAssign. The box encoding method adopts APE, which is similar to rfpoint.

of 17.2 frames/s, and the mAP of 73.20% with a speed of 87 frames/s corresponding to the experiment parameters of FCOSR‡ [15] and GGHL‡ [14]. Similar to FCOSR [15] and GGHL [14], the proposed method also focuses on FCOS [13] and performs better than the two on precision and inference speed. Compared with these methods, the propose method eliminates the centerness head and adopts simple oriented box representation forms, promoting the increase in inference speed. Meanwhile, an effective sample definition strategy ensures detection precision. Treating S^2A-Net‡ [3] as the benchmark, the mAP of the proposed method reaches 77.74%, lower than 79.42% of S^2A-Net‡ [3]. However, S^2A-Net [3] is an anchor-based detector and contains several modules to align rotating features. The proposed method is an anchor-free detector and only relies on a dynamic loss function equilibrium to choose proper features. It needs strong feature extractors to bridge the gap when training data are complex. With basic augmentation and single-scale training, the mAP of the proposed method reaches 73.47%, close to 74.12% of S^2A-Net [3]. Considering inference speed, the inference speed of S^2A-Net [3] is 16 frames/s, while that of the proposed method reaches 24 frames/s.

D. Comparisons With the State of the Art

The performance between the proposed method and other state-of-art detectors on DOTA, DIOR, and HRSC2016 is comprehensively compared in this section.

1) Results on the DOTA Dataset: DOTA is the most commonly used dataset benchmark in aerial image detection, and numerous methods experimenting on this dataset are available. The performance difference between the proposed method and some recent mainstream methods (or related methods) is compared, and the results are listed in Table VI in detail. Results show that the proposed method is effective in almost all categories and reaches the same level as recent mainstream methods [3], [4], [10], [12], [14], [15], [18], [22], [23], [25], [31], [34], [37], [38]. These mainstream methods mainly comprise anchor-based detectors. Among them, RoI Trans [1] and gliding vertex [4] are two-stage detectors, while FCOSR [15] and AutoAssign-O [10] are one-stage detectors. Without dense preset anchors, the proposed method still achieves good performance based on one-stage detector structures, especially for those targets with large aspect ratios. On LV, ships, and harbors, the proposed method separately obtains the mAP of 85.58%, 88.83%, and 78.62%, which are remarkably competitive among these methods. Fig. 10 shows visualization comparisons among FCOSR [15], AutoAssign-O [10], and the proposed method. These results also prove that the absence or presence of dense anchors is not the key factor affecting the performance of large-aspect-ratio targets.

2) Results on the DIOR Dataset: The proposed method is tested on the DIOR dataset and compared with several mainstream detectors whose baselines are provided by DIOR. The visualizations of proposed method are shown in Fig. 11. Experiments in Table VII show that the proposed method performs substantially better than other compared methods on a portion of the categories and reaches a high indicator of mAP. On categories of airport (APO), bridge (BR), chimney (CH), dam (DAM), GF, overpass (OP), ship (SH), storage tank (STO), and vehicle (VE), at least a 0.41% increase is observed, and the highest increase can reach almost 13.32% on some categories when compared with other state-of-the-art detectors using ResNet50. The proposed method also achieves competitive performance in other categories. Considering the

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TABLE VI

COMPARISON OF THE PROPOSED METHOD WITH THE PERFORMANCE OF OTHER DETECTORS ON THE DOTA DATASET. † INDICATES TRAINING WITH EXTRA RANDOM ROTATION AS DATA AUGMENTATION. ‡ DENOTES MULTISCALE TRAINING AND TESTING ON THE BASIS OF †.

Method	Backbone	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP
two-stage:																	
RoI Trans.‡ [1]	R101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
Gliding Vertex† [4]	R101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
AOPG [34]	R101	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.6	66.3	74.05	67.76	58.77	75.39
APE† [18]	RX101	**89.96**	83.62	53.42	76.03	74.01	77.16	79.45	90.83	87.15	84.51	67.72	60.33	74.61	71.84	65.55	75.75
Oriented R-CNN [37]	R101	89.46	82.12	54.78	70.86	78.93	83	88.2	**90.9**	87.5	84.68	63.97	67.69	74.94	68.84	52.28	75.87
CenterMap‡ [23]	R101	89.83	84.41	54.6	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
ReDet [38]	ReD50	88.79	82.64	53.97	74	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
one-stage:																	
Axis-Learning [5]	R101	79.53	77.15	38.59	61.15	67.53	70.49	76.3	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05	65.98
A^2S-Det† [11]	R101	89.59	77.89	46.37	56.47	75.86	74.83	86.07	90.58	81.09	83.71	50.21	60.94	65.29	69.77	50.93	70.64
R^3Det‡ [2]	R50	89.09	84.13	55.77	74.48	77.71	82.99	87.57	89.46	84.89	85.67	66.09	64.17	75.13	75.35	62.78	77.02
AutoAssign-O [10]	R50	89.23	77.56	49.72	75.6	76.76	75.39	83.47	90.87	82.75	82.38	57.86	66.26	65.79	68.69	45.02	72.49
S^2A-Net [3]	R50	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.9	85.64	60.36	62.6	65.26	69.13	57.94	74.12
S^2A-Net‡ [3]	R50	88.89	83.6	57.74	81.95	79.94	83.19	**89.11**	90.78	84.87	87.81	70.3	68.25	78.3	77.01	69.58	79.42
S^2A-Net+DAL‡ [12]	R50	89.69	83.11	55.03	71	78.3	81.9	88.46	90.89	84.97	87.46	64.41	65.65	76.86	72.09	64.35	76.95
R^3Det+GWD‡ [22]	R50	88.43	84.33	56.91	**82.19**	76.69	83.23	86.78	88.9	83.93	85.73	72.07	65.67	76.76	78.37	65.31	78.35
FSDet† [31]	R50	89.2	84.33	55.38	74.81	80.85	78.79	88.24	90.85	86.85	85.85	67.14	64.79	76.49	75.11	69.02	77.85
FoRDet‡ [25]	VGG16	89.62	**85.88**	47.55	81.45	80.63	81.84	88.08	90.87	**88.27**	86.41	**72.42**	67.69	73.91	72.67	64.63	78.13
GGHL‡ [14]	DarkNet53	89.74	85.63	44.5	77.48	76.72	80.45	86.16	90.83	88.18	86.25	67.07	69.4	73.38	68.45	70.14	76.95
FCOSR‡ [15]	RX50	89.06	84.93	52.81	76.32	81.54	81.81	88.27	90.86	85.20	87.58	68.63	70.38	75.95	**79.73**	75.67	79.25
rfpoint (ours)	R50	88.6	79.39	50.69	70.06	78.71	79.35	87.63	90.89	82.72	84.75	58.61	65.84	66	69.2	49.64	73.47
rfpoint (ours)	R101	88.72	79.49	50.01	69.22	78.6	80.31	87.43	90.88	81.89	84.72	57.78	62.85	66.97	69.47	47.15	73.03
rfpoint (ours)	Swin-t	88.87	83.94	50.85	71.61	80.55	83.67	88.24	90.9	86.73	86.88	61.94	66	68.05	73.72	57.51	75.96
rfpoint‡ (ours)	R50	88.88	82.15	56.39	72.91	80.64	85.16	88.67	90.78	84.27	87.9	64.12	61.55	78.06	75.11	69.55	77.74
rfpoint‡ (ours)	R101	89.25	83.42	55.23	76.84	81.18	85.23	88.86	90.84	86.55	**88.44**	66.73	66.25	78.03	77.41	67.49	78.78
rfpoint‡ (ours)	Swin-t	88.64	85.64	**59.77**	78.76	**81.66**	**85.58**	88.83	90.88	83.85	87.1	69.84	67.3	**78.62**	76.28	**76.12**	**79.92**

TABLE VII

COMPARISON OF THE PROPOSED METHOD WITH THE PERFORMANCE OF OTHER DETECTORS ON THE DIOR DATASET. THE RESULTS OF OTHER DETECTORS ARE OBTAINED FROM THE BENCHMARK PROVIDED IN DIOR [34]. NO EXTRA RANDOM ROTATION IS ADOPTED AS DATA AUGMENTATION

Method	Backbone	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
Faster RCNN-O [39]	R50	62.79	26.8	71.72	80.91	34.2	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
RetinaNet-O [32]	R50	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.2	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
GLIDING VERTEX [4]	R50	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.7	72.3	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
RoI Trans. [1]	R50	63.34	37.88	71.78	87.53	40.68	72.6	26.86	**78.71**	68.09	68.96	**82.74**	**47.71**	55.61	81.21	**78.23**	70.26	81.61	54.86	43.27	65.52	63.87
AOPG [34]	R50	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	**69.99**	64.41
rfpoint (ours)	R50	63.48	51.2	69.68	86.55	42.92	75.06	31.82	68.46	74.11	77.52	76.54	41.76	56.67	87.62	64.42	71.79	81.61	55.83	52.79	66.18	64.8
rfpoint (ours)	Swin-t	**77.75**	**59.5**	**79.31**	**89.43**	**45.52**	**80.87**	**40.89**	76.00	**84.1**	**83.72**	82.01	43.95	**60.62**	**88.59**	70.71	**78.56**	**86.67**	**66.46**	**53.87**	69.16	**70.88**

entire performance, the mAP of the proposed method reaches 64.8%, which is higher than 60.06% of gliding vertex, 63.87% of RoI Trans [1], and 64.41% of anchor-free oriented proposal generator (APOG) [34]. With the backbone of Swin-t, the proposed method reaches the best performance of 70.88%. These mainstream detectors belong to two-stage detectors, while the proposed method is a one-stage detector with an anchor-free pattern.

3) Results on the HRSC2016 Dataset: Results on HRSC2016 are evaluated under the PASCAL VOC2007 and PASCAL VOC2012 metrics, where evaluation IoU thresholds are 0.5 and 0.75 represented as mAP_{50} and mAP_{75}, respectively. As shown in Table VIII, the proposed method achieves the best performance among all compared methods [1], [3], [4], [12], [15], [22], [23], [25], [31], [34], [38] regardless of its evaluation under VOC_{07} metric or VOC_{12} metric. Despite reaching the best mAP_{50} on HRSC2016 (90.67% for VOC_{07} metric and 98.23% for VOC_{12} metric), the accuracy gaps of the proposed method with other detectors are still unclear. Compared with S^2A-Net [3], GWD [22], and FCOSR [15], the increases are only (0.02%, 1.07%), (0.82%, 0.86%), and (0.53%, 2.49%) under evaluation metrics of (VOC_{07}, VOC_{12}), which reach the limitation of mAP_{50} on HRSC2016. The indicator of mAP_{75}, which can strictly evaluate the performance of box prediction, is also calculated to analyze the superiority of the proposed method on HRSC2016 further. The mAP_{75} of 88.83% reflects the precise prediction of the bounding box of aerial targets by the proposed method, while mAP_{75} of DAL [12] and GWD [22] is only 67.6% and 65.88%, respectively. A method that provides mAP_{75} under VOC_{12} metric is currently unavailable. However, the performance of mAP_{75} under VOC_{12} metric is still provided for future comparison.

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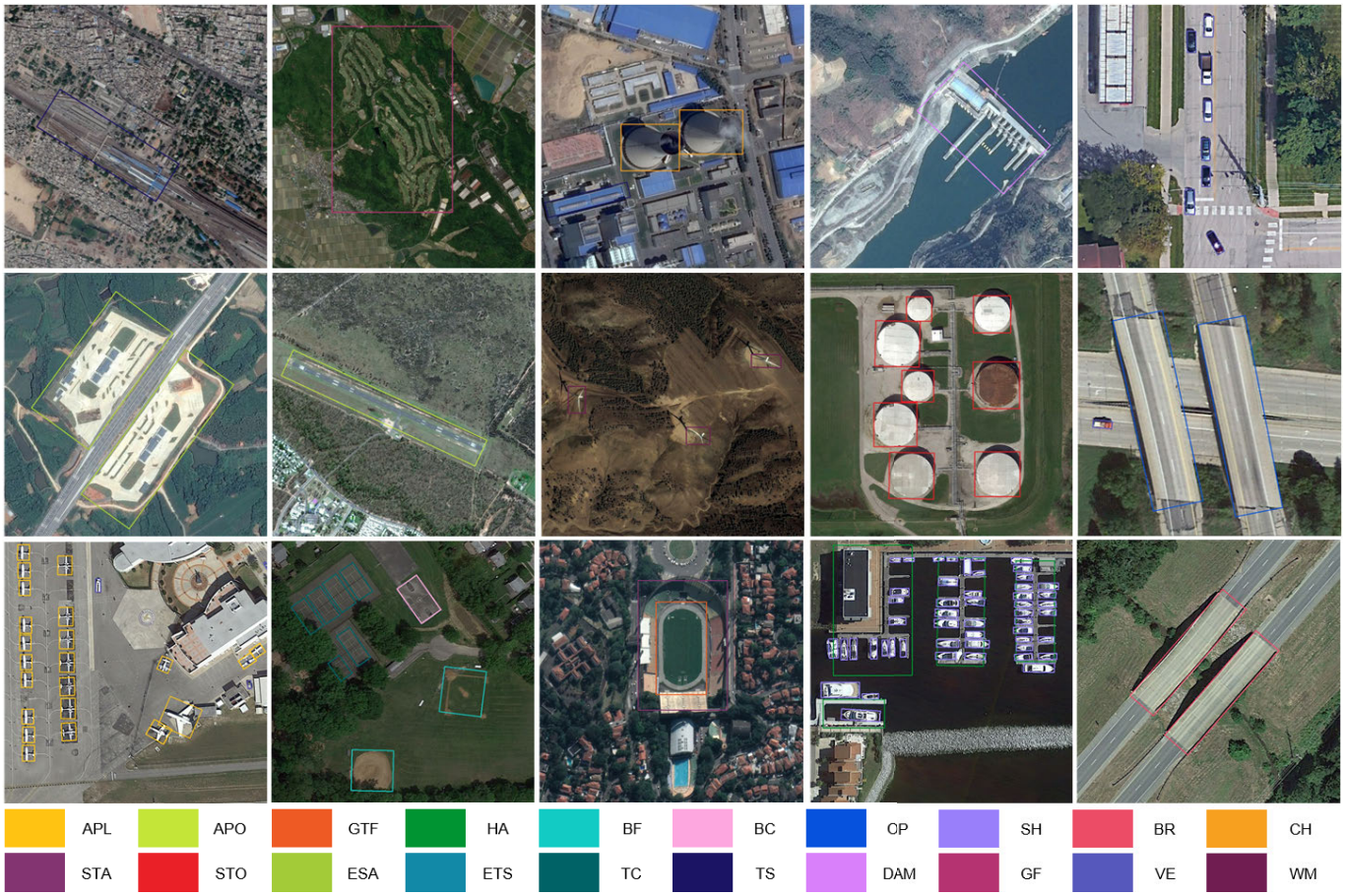


Fig. 11. Visualizations of predictions using the proposed method on the DIOR dataset.

TABLE VIII

COMPARISON OF THE PROPOSED METHOD WITH THE PERFORMANCE OF OTHER DETECTORS ON THE HRSC2016 DATASET. * REPRESENTS ACCURACIES EVALUATED BY mAP_{75}, OTHERWISE EVALUATED BY mAP_{50}. † DENOTES TRAINING WITH EXTRA RANDOM ROTATION AS DATA AUGMENT

Methods	Backbone	Image Size	mAP (07)	mAP (12)
RoI Trans. [1]	R101	512×800	86.2	-
Gliding Vertex [4]	R101	512×800	88.2	-
AOPG [34]	R101	800×1333	90.34	96.22
CenterMap [23]	R101	-	-	92.8
ReDet [38]	Re50	512×800	90.46	97.63
S^2A-Net [3]	R101	512×800	90.17	95.01
S2A-Net+DAL† [12]	R50	800×800	88.60 / 67.6	-
R3Det-GWD† [22]	R101	512×512	89.85 / 65.88	97.37
FoRDet† [25]	VGG16	512×512	89.92	-
FSDet† [31]	R101	800×800	90.38	-
FCOSR† [15]	X101	800×800	90.14	95.74
rfpoint (ours)	R50	800×800	90.21 / 79.85*	95.28 / 83.26*
rfpoint (ours)	R101	800×800	90.19 / 80.23*	96.08 / 85.54*
rfpoint (ours)	Swin-t	800×800	90.15 / 78.66*	95.87 / 80.18*
rfpoint† (ours)	R101	800×800	**90.67 / 88.83***	**98.23 / 89.26***

Overall, under a remarkably challenging evaluation indicator, the proposed method still achieves the best performance and is far more effective than other methods on HRSC2016. Moreover, visualizations of predictions shown in Fig. 12 prove the proposed detector's effectiveness on HRSC2016.



Fig. 12. Visualizations of predictions using the proposed method on the HRSC2016 dataset.

V. CONCLUSION

The limitations of anchor-free oriented detectors are analyzed in this study to balance speed and precision. To address these limitations, an anchor-free oriented one-stage detector (termed rfpoint), which requires minimal prior knowledge and can model sample definition as a dynamic regulating process, is proposed. Moreover, RQDL and AWBL, which interact until model convergence, are designed. In this process, the positive and negative properties of training samples are constantly changing. Several experiments on three mainstream datasets were also designed, achieving the best performances in the DIOR and HRSC2016 datasets and competitive performance in the DOTA dataset. For the inference speed, rfpoint achieves the best speed compared with other rotating detectors. However, some qualitative explorations are conducted, and some imperfections may be observed. With almost no speed decay,

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label assignment provides remarkable contributions to the performance gains of fast rotating detectors and increases its competitiveness with other anchor-based oriented detectors. Future work will aim to explore the quantitative effects of label assignment and exploit the full potential of fast oriented detectors.

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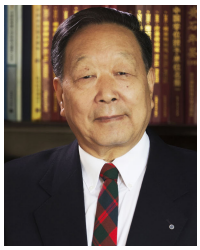
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